

CREDIT CARD FRAUD DETECTION ANALYSIS REPORT

Executive Summary

This project presents an end-to-end Credit Card Fraud Detection System developed to identify fraudulent transactions within a highly imbalanced financial dataset containing 284,807 transactions. Fraudulent transactions represented only 492 records (0.17%), creating a significant machine learning challenge. The project combined Exploratory Data Analysis (EDA), class imbalance handling using SMOTE, supervised learning through XGBoost, and anomaly detection using Isolation Forest. The final solution achieved high fraud detection performance while minimizing false alarms and was deployed through an interactive Streamlit dashboard for real-time monitoring and prediction.

Introduction

Financial fraud continues to be one of the most critical challenges faced by banking and payment institutions. Traditional rule-based systems often struggle to detect evolving fraud patterns. This project was designed to build a data-driven fraud detection framework capable of identifying suspicious transactions and supporting proactive fraud prevention strategies.

Background

The dataset consists of anonymized credit card transactions made by European cardholders. Due to confidentiality requirements, most features were transformed using Principal Component Analysis (PCA). The primary challenge was the extreme class imbalance, where legitimate transactions significantly outnumbered fraudulent ones. Such imbalance can cause conventional machine learning models to overlook fraud cases and produce misleading accuracy scores.

Purpose

The objective of this project was to:

- Detect fraudulent transactions with high precision and recall.
- Address severe class imbalance through advanced resampling techniques.
- Compare supervised and unsupervised machine learning approaches.
- Develop a practical fraud monitoring solution suitable for real-world deployment.
- Provide business insights that can help reduce financial losses and improve transaction security.

Dataset Overview

Total Transactions: 284,807

Fraudulent Transactions: 492

Legitimate Transactions: 284,315

Fraud Rate: 0.1727%

Features: 30 Predictor Variables + Target Variable

Missing Values: 0

Duplicate Records Removed During Preprocessing

Methodology

1. Performed Exploratory Data Analysis to identify transaction patterns, outliers, and fraud distribution.
2. Applied Standard Scaling to transaction amount and time features.
3. Used SMOTE (Synthetic Minority Oversampling Technique) to balance the minority fraud class.
4. Developed and evaluated three models:
5. Logistic Regression (Baseline)
6. XGBoost Classifier
7. Isolation Forest (Anomaly Detection)
8. Compared models using Precision, Recall, F1 Score, ROC-AUC, and Confusion Matrices.
9. Built a real-time fraud detection dashboard using Streamlit.

Key Findings & Analysis Impact

- Fraudulent transactions accounted for less than 0.2% of the dataset, confirming the need for specialized imbalance-handling techniques.
- Several PCA-transformed variables showed strong relationships with fraudulent activity, indicating distinct behavioral patterns between fraudulent and legitimate transactions.

- XGBoost significantly outperformed the baseline model by achieving a stronger balance between fraud detection capability and false positive reduction.
- Isolation Forest successfully identified anomalous transactions without requiring labeled data, demonstrating its usefulness for detecting emerging fraud patterns.
- The hybrid approach increased robustness by combining known fraud detection with anomaly-based identification.

Model Comparison Summary

Model	Business Value
Logistic Regression	Baseline benchmark and interpretability
Isolation Forest	Detection of unseen fraud patterns
XGBoost	Highest overall fraud detection performance

XGBoost was selected as the final production model due to its superior Precision, Recall, F1 Score, and ROC-AUC performance.

Business Impact

The developed fraud detection system can provide the following organizational benefits:

- Faster identification of suspicious transactions.
- Reduced financial losses caused by fraudulent activities.
- Improved customer trust through enhanced transaction security.
- Lower operational workload by reducing false fraud alerts.
- Better risk management and decision-making through data-driven monitoring.

If deployed at scale, even a small improvement in fraud detection accuracy could potentially prevent thousands of dollars in fraudulent losses and significantly improve operational efficiency.

Recommendations

- Implement real-time transaction scoring using the trained XGBoost model.
- Integrate automated alert systems for high-risk transactions.
- Continuously retrain models with new transaction data to adapt to evolving fraud behaviors.

- Incorporate graph analytics and deep learning methods in future versions to improve fraud network detection.

Conclusion

This project successfully demonstrated how machine learning and anomaly detection techniques can be leveraged to address highly imbalanced fraud detection problems. Through comprehensive data analysis, model comparison, and dashboard deployment, the solution provides a practical framework for detecting fraudulent transactions and supporting proactive financial risk management.